

NERIS: The New National Fire Incident Data Standard

Opportunities for the Actuarial & Insurance Community — Replacing NFIRS (1973–2025)

PROJECT CRUCIBLE

Prepared for:

Arizona Captive Insurance Association
Casualty Actuaries of the Desert States

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NFIRS — the 40-year-old national fire reporting system — was decommissioned in early 2026. NERIS, its modern successor, is now live: standardized, near-real-time, geospatial fire incident data for ~27,000 U.S. fire & EMS departments.

~27,000

Fire & EMS Depts. Targeted

<24 hrs

Data Lag (was 6–18 months)

3

Incident Types per Record (vs. 1 in NFIRS)

\$0

Cost to Fire Depts. & Verified Research Orgs

WHAT IS NERIS?

The **National Emergency Response Information System** is a cloud-based, all-hazards incident reporting platform built by the **Fire Safety Research Institute (FSRI / UL Research Institutes)** with the **U.S. Fire Administration (USFA)** and **DHS S&T Directorate**.

NERIS replaced NFIRS on **January 1, 2026**.

First live incident: January 2025. Nationwide rollout completed across all FEMA regions through 2025.

Governance: Departments own their data. The platform aggregates and anonymizes for national publication — no PII in public feeds.

KEY DATA FIELDS CAPTURED

- ✓ Incident type (multi-tag: fire, EMS, hazmat, WUI, etc.)
- ✓ Geospatial coordinates (GIS-native lat/lon)
- ✓ Location type & property use classification
- ✓ Dispatch, en-route, arrival & clear timestamps
- ✓ Suppression tactics & actions taken
- ✓ Civilian & firefighter casualties / rescues
- ✓ Property loss & contents loss (structured)
- ✓ Detector / suppression system performance
- ✓ CAD/dispatch pre-population (automated)

Schema: github.com/ulfsri/neris-framework (public)

NFIRS VS. NERIS

Attribute	NFIRS	NERIS
Data lag	6–18 months	<24 hours
Incident types	Single code	Up to 3 tags
Geospatial	Address only	GIS-native
API access	None	RESTful (public)
Property loss	Ranges	Structured fields
Response times	Limited	Granular timestamps
Format	Numeric codes	Plain-language
Public data	Annual bulk	API + near-real-time

TWO CLASSES OF OPPORTUNITY FOR INSURANCE & ACTUARIAL PROFESSIONALS

CLASS 1 Cooperative Data Partnerships — Direct access to department-level (non-aggregated) data via data-use agreements with departments, State Fire Marshals, or FSRI/USFA:

- Hyper-local response time analysis for property underwriting
- Suppression system effectiveness for commercial risk classification
- WUI incident density overlaid with policy portfolios
- ISO/PPC rating validation and improvement support

CLASS 2 Public Aggregated / API Feed — Free, PII-free, near-real-time national data available to verified organizations:

- Dynamic ISO score modeling using actual incident & response metrics
- Community Risk Reduction (CRR) data for loss-prevention credits
- Statewide trend analysis for rate-making & catastrophe modeling
- Academic research partnerships (SOA, CAS curriculum)

DATA ACCESS PATHWAYS

Channel	Who	Cost
NERIS Public API	Verified orgs	Free
Aggregated Download	Public/researchers	Free
Dept.-level DUA	Negotiated	TBD
State Fire Marshal	State-level	Varies
NFIRS Historical	OpenFEMA	Free

ISO / PPC Link: Fire department reporting completeness and response performance data in NERIS directly factors into ISO Public Protection Classification (PPC) scores — the metric driving homeowner and commercial property premiums. Better data = demonstrable PPC improvement.

Historical Baseline: 45 years of NFIRS data (1980–2025) on OpenFEMA for retrospective analysis.

PROPOSED PILOT: PROJECT CRUCIBLE — ARIZONA FIRE RISK DATA PIPELINE

Phase 1 — Connect: Establish API integration with NERIS aggregated feed for Arizona incident data. Map NERIS geospatial fields to existing property/policy databases.

Phase 2 — Analyze: Apply ISO 31010 risk methodology to response time, suppression performance, and property-loss fields. Identify underwriting gaps in WUI and high-risk corridors.

Phase 3 — Underwrite: Deliver dynamic risk scoring to captive and standard market carriers — enabling accurate rating of previously uninsurable fire-risk properties.

Sources: USFA/FEMA (usfa.fema.gov/nfirs/neris) · FSRI/UL Research Institutes (neris.fsri.org) · NERIS Schema (github.com/ulfsri/neris-framework)

NC OSFM · NY DHSES · Nebraska State Fire Marshal · Texas Dept. of Insurance · International Fire & Safety Journal (Jan 2026)

Richards+ / Linked Trust / AI Trailblazers / University of Arizona Risk Runners

jefferson@richards.plus · 520.981.3639 · richards.plus